

PROJECT SUTRA — GRAND FINALS ORAL SCRIPT

WORD-FOR-WORD DEFENSE & DISASTER ROBOTICS JURY PRESENTATION
(SLIDES 1 TO 3)

TOTAL TIME: 3 MIN 00 SEC
TARGET AUDIENCE: JURY / NDMA /
DRDO

SLIDE 01 Title & Vision: Decentralized Swarm Reconnaissance

00:00 - 00:45 (45s)

SPEAKER: NIKHIL (TECH LEAD)

TONE: AUTHORITATIVE, GROUNDED, TACTICAL

"Respected jury members, disaster response leaders, and fellow engineers.

When a mountain slope collapses in Wayanad or flash floods rip through Sikkim, traditional disaster management hits a complete wall. Satellite GPS signals are blocked by thick tree canopies, cellular base towers are washed away, and high-bandwidth video links crash behind mountain ridgelines.

We are **Team OFFGRID**, and we have engineered **PROJECT SUTRA** — the *Swarm Unified Tactical Reconnaissance Architecture*.

SUTRA is an autonomous, decentralized multi-UAV swarm system built to find survivors where conventional robotics fail. By integrating **50Hz PX4 MicroXRCE decentralized offboard flight**, **Deep JSCC neural semantic video** that operates down to -5dB SNR without digital cliff dropouts, and **3.59cm real-time WGS84 3D DEM terrain raycasting**, SUTRA transforms disaster search from a chaotic 3-hour manual operation into an autonomous 10-minute sweep."

[VISUAL CUE]: Point to the 3 grounded telemetry tabs (50Hz GNC, -5dB Deep JSCC, 3.59cm Elevation Accuracy) and introduce the 5-member engineering team leads (Nikhil, Vedanth, Siva, Harika, Rohith).

SPEAKER: NIKHIL → TRANSITION TO VEDANTH & SIVA TONE: URGENT, EMPIRICALLY RIGOROUS, FACT-DRIVEN

"To understand why search-and-rescue fails today, our team audited actual NDRF and SDRF disaster after-action reports across 4 foundational engineering voids:

- 1. Subsystem A Void (Navigation & GNC):** Under dense forest canopies, GNSS satellite multi-path causes commercial flight controllers to drift and crash. In Wayanad, over **70% of standard canopy sorties resulted in airframe rotor losses**.
- 2. Subsystem B Void (Comms & Video):** Standard H.264 digital video suffers from the brutal '*Digital Cliff*'. The moment a drone dips behind a mountain ridge and RF signal-to-noise ratio drops below 5dB, video packets completely corrupt, leaving rescue commanders blind.
- 3. Subsystem C Void (AI Vision & Geolocation):** Current drone AI runs 2D bounding boxes that assume flat, zero-elevation ground. On 45-degree Himalayan mountain slopes, flat raycasting produces **15 to 30 meters of calculation error**, routing rescue squads to empty cliffs.
- 4. Subsystem D Void (Tactical C2):** Traditional setups require 1 pilot per drone and a massive **15 to 25 person ground crew**. If the single pilot link drops, the entire sortie aborts — costing up to **₹12.5 Lakhs per deployment** while burning the golden 72-hour survivor window."

[VISUAL CUE]: Emphasize the 4 embedded carousel disaster evidence photos (Wayanad Canopy Crash, Chamoli Gorge RF Cut, Sikkim Slope Geo Error, and Central Link Abort).

SPEAKER: NIKHIL (CLOSING WITH BENCHMARKS & MARKET DEFENSE) TONE: TRIUMPHANT, DEFENSIBLE, COMMERCIAL READINESS

"Project SUTRA directly answers all 4 problem voids through 4 hardened technological moats:

- **Moat A — SUTRA-FSD & 3D ORCA:** Decentralized 3D velocity obstacle avoidance running at 50Hz offboard over MicroXRCE-DDS with zero reliance on ground control towers or external GNSS locks.
- **Moat B — Deep JSCC Neural Video:** Differentiable joint source-channel coding mapped directly to analog channel symbols. When RF SNR drops down to -5dB in deep ravines, video degrades gracefully into clean silhouettes with zero blackouts.
- **Moat C — 3D DEM Real-Time Geolocation:** Edge-accelerated YOLOv8-Pose on NVIDIA Jetson Orin coupled with real-time WGS84 Digital Elevation Model raycasting, delivering pinpoint **3.59cm survivor geolocation** in under 15ms.
- **Moat D — Pegasus 3D WebGPU GCS:** An offline tactical digital twin and ATAK plug-in that enables **1 to 2 operators to manage a 6-drone swarm** at a total sortie cost of just **₹95,000** — an 85% cost reduction.

With a global disaster robotics market expanding to **₹74,500 Crores** and **₹1,500 Crores allocated under NDMA/SDRF** in 2025, SUTRA is engineered, benchmarked, and mission-ready to save lives."

Metric Benchmark	Traditional SAR	Single Drone	PROJECT SUTRA SWARM
Deployment Speed	45-90 min setup	15-30 min	8-15 min (Autonomous Launch)
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (3-4x Faster)
Survivor Detection Rate	65-70%	55-65%	78-85% First-Pass Detection
Crew Personnel Required	15-25 specialists	2-3 pilots	1-2 Tactical Operators
Cost per Deployment	₹12.5 Lakhs (\$15K)	₹5.0 Lakhs	₹95,000 (\$1.1K) Sortie